

Fractional-Order Dynamics of Cardio-Respiratory Oxygen Transport: A Stable Regulator of Human Endurance and the Boundary of Chaos

Aashima Bangia¹, Rashmi Bhardwaj², Asokan Vasudevan³, Naeema Darwish Khamis Al Maashari⁴, Yafang Yan⁵, Khin Yadanar Htun⁵, Email: ORCID:⁵, Email:⁵, Email: ORCID:⁶, AMS Classification: 37M⁵, 37M⁵, 37Fxx⁵, 65-XX⁵, Physiological basis of the model assumptions⁵, Selection of parameter values⁵, leading to⁵, f can be approximated through Jacobian matrix as:⁵, For the system under study⁵, the jacobian⁵, J can be intended as:⁵, Consecutively⁵, Characteristic Polynomial⁵, $P(n)$ so formed is:⁵, Definition: Grunwald-Letnikov Derivative⁵, Binomial coefficient present in the derivative can be computed through Gamma function as:⁵, Theorem: Riemann Liouville Fractional Derivative (RLFD)⁵, Riemann alongwith Liouville in the year 1847 developed this derivative as:⁵, having as the well-known gamma function computed through:⁵, Now⁵, employing fractionalization algo for extending this fractional equation i.e.⁵, as⁵, the⁵, Whenever⁵, the RL operator reduces the equation to:⁵, This is equivalent to the reduced RLFD given in equation (3).⁵, Definition: Caputo Derivative for Fractional order⁵, Caputo derivative is defined for fractional order⁵, as:⁵, Definition: Riemann-Liouville Fractional Integral(RLFI)⁵, Definition: Caputo Fractional Integral⁵, The integral for the order⁵, defined in the sense of Caputo as:⁵, having as the well-known gamma function computed through:⁵, having initial conditions:⁵, where denotes the operator as defined by the Caputo as:⁵, Consider the integer order general form of differential equations represented as:⁵, having initial conditions:⁵, which leads to the integration results derived given in [] as follows:⁵, Rearranging the integration eqn. (16) as follows:⁵, On employing trapezoidal rule on eqn. (16) alongwith (17)⁵, it becomes⁵, then eqn. (16) becomes: such as ⁵, Thus⁵, Therefore⁵, eqn. (20) in terms of eqn. (22) becomes:⁵, And the predictor form⁵, of this fractional Adams-Moulton method can be seen as:⁵, This implies that as per the Adams-Moulton method is represented as:⁵, Phase⁵, Variation in $q = [q^5, q^5, q^5]$ ⁵, Variation in time period⁵, t^5 , Stable⁵, Critical (extended model)⁵, Chaotic (extended model)⁵, Fractional order⁵, q^5 , $-\arg \lambda - \min(\text{rad})^5$, $q\pi^5$, Regime⁵, Yes⁵, Stable focus⁵, Yes⁵, Stable focus⁵, Yes⁵, Stable focus⁵, Yes⁵, Stable focus⁵, Yes⁵, Stable focus⁵, Yes⁵, Stable focus⁵, Yes⁵, Stable focus⁵, Table-2: Values of parameters considered in the model⁵, Parameter⁵, Description⁵, Values⁵, M^5 , Rate of Metabolism⁵, 0.01/sec⁵, V^5 , Maximum lung volume⁵, 3600ml⁵, Maximum transmission rate of molecules from lungs to bloodstream⁵, 0.1/sec⁵, Maximum transmission rate of molecules from bloodstream to bodycells⁵, 0.7/second (⁵), Breathing-period⁵, 20s inhale⁵, 20s suspension⁵, 20s exhale i.e.⁵, 60 secs in total⁵, Amount of oxygen intake under different circumstances⁵, 24L/minute (⁵), Table 3: Literature studies for the kinesiology-based models⁵, Year⁵, Author⁵, Description⁵, Parameters⁵, Results⁵, Complexity Dynamics of Meditating Body⁵, Differential model developed for the human body⁵, Analysis of the human body during different meditation phases to capture varied oxygen concentration-levels⁵, Teka W.W. et al⁵, Spiking as well as overflowing patterns of fractional-order Izhikevich model⁵, Fractional order equations developed for Izhikevich model⁵, Analyzed the parameters of the fractional order apt for simulation⁵, Dynamical Indicator of Human Body's

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Physical Endurance⁵, Fractional ordered SEQIR-based model was developed to study Coronavirus dynamics⁵, Numerical simulations plotted⁵, Lyapunov stability found the system to be asymptotically stable⁵, Ghita M.⁵, Copot D.⁵, Ionescu C.M.⁵, Lung cancer dynamics using fractional order impedance modeling on a mimicked lung tumor setup⁵, Fractional order impedance modelling was developed to study lung tumor⁵, Li Z⁵, Pei Y⁵, Wang Y⁵, Tian Q⁵, An enhanced respiratory mechanics model based on double-exponential⁵, fractional calculus⁵, Fractional respiratory mechanics fitted to real data⁵, The hybrid model achieved significantly better results for different ventilation modes⁵, Alharbi. N⁵, Present study⁵, Fractional Ordered Kinesiology Model⁵

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⁵ This derivative is used to describe the models with complexities as RLFD becomes incompetent to deal with the intricate systems.

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Abstract: This work develops a fractional-order dynamical model of coupled cardio-respiratory oxygen transport among the lungs, the heart and bloodstream, and the body tissues, and studies the conditions under which the human body sustains or loses stable oxygen regulation during sustained effort. The three-compartment system is formulated with Caputo fractional derivatives of order q , so that physiological memory arising from delayed oxygen uptake, circulatory transport lags, and gradual tissue utilisation is represented intrinsically rather than through added delay terms. The equilibrium structure is derived, and the associated Jacobian matrix is obtained in closed form as a real, non-symmetric operator. Local stability is established rigorously using the Matignon criterion appropriate to fractional-order systems: at the physiological operating point the eigenvalues give a minimum argument of 1.605 radians, yielding a critical order of 1.02, so the regulated state is a stable focus for every admissible order between zero and one. This analytical result is confirmed numerically by the largest Lyapunov exponent, computed with a variational method validated on the classical Lorenz system, which returns a value of -0.10 independent of the initial condition. The model therefore describes a robustly self-regulating oxygen-transport system, and this stable regime is interpreted as the dynamical basis of endurance. A physiologically motivated extension in which the cardiac response gain is separated from and exceeds the pulmonary transfer coefficient and the metabolic dissipation is raised to realistic levels undergoes a genuine transition to chaotic dynamics, offering a candidate description of cardio-respiratory dysregulation and fatigue. Notably, stronger memory (a lower fractional order) suppresses this chaos, indicating that the fractional formulation enhances the robustness of the regulated endurance state. A parameter-sensitivity analysis confirms that stability persists across the physiological ranges of the transport, breathing, and metabolic parameters. The findings connect fractional-order nonlinear dynamics to measurable endurance and recovery in sports science, physiotherapy, and cardio-respiratory health monitoring. Numerical simulations employ a fractional Adams–Bashforth–Moulton (ABM) scheme.

Keywords: Caputo fractional derivative, Cardio-respiratory coupling, Chaos, Largest Lyapunov exponent, Oxygen transport, Fractional order Differential Equations (FODE), Fractional order Modelling, Adam Bashforth Moulton (ABM)

Conclusions

From this case-study, most importantly the relation between physical activities, exercises, postures health and happiness can be seen as mutually exclusively through important components such as Heart, Lungs and Cells/Tissues in the body is studied using Lyapunov and phase space analysis numerically simulated on MATLAB. The nonlinear differential model determines change in concentration for oxygen during musculoskeletal physical activities based on two major components heart and energy utilized by the Adenosine triphosphate (ATP) molecules using the compartment model of breath function. Model utilizes non-linear model equalities which are with respect to time at constant rate of metabolism. The phase diagram of the setup has been analyzed for the behavioral study of the equilibrium (fixed) point with the nature of that point. Eigen-system leads to phase space diagram studied the regular and chaotic behavior. The fractional order values of $q=[q_1, q_2, q_3]$ for the 'LHB' model vary from $0 < q < 0.25$ for stable case, $0.25 < q < 0.5$ for critical case and have values $0.5 < q < 1$ for the chaotic phase. Thus, it is recommended that for extensive exercising and to avoid collapsing, primarily the breath function should be boosted so that the person can hold their breath for a comparatively longer period. This will in turn boost the stamina as more and more of ATP molecules will be formed and utilized.

In practical terms, the stable regulated regime provides a candidate quantitative marker of endurance: the stability margin of the oxygen-transport dynamics could be used to characterise cardio-respiratory fitness, to monitor recovery in physiotherapy, and to flag early dysregulation in health monitoring. The principal limitations are the single-subject parameterisation, the reliance on literature-based rather than individually measured parameters, and the idealised three-compartment structure; the chaotic extension, though physiologically motivated, remains to be tested against clinical data. Future work will pursue individual calibration against measured oxygen-uptake data, extension to time-varying workload, and experimental testing of the predicted stabilising role of physiological memory.

Acknowledgement

$$y_{k+1}^p = \sum_{\mu=0}^{n-1} y^{(\mu)}(0) \frac{t_{k+1}^\mu}{\mu!} \frac{1}{\Gamma[\lambda]} \left[\sum_{\mu=0}^k \beta_{\mu,k+1} g(t_\mu, y_\mu) \right]$$

$$y_{k+1} = \sum_{\mu=0}^{n-1} y^{(\mu)}(0) \frac{t_{k+1}^\mu}{\mu!} \frac{1}{\Gamma[\lambda]} \left[\sum_{\mu=0}^k \alpha_{\mu,k+1} g(t_\mu, y_\mu) + \alpha_{\mu+1,k+1} g(t_{\mu+1}, y_{\mu+1}^p) \right]$$

$$y_{k+1}^p = \sum_{\mu=0}^{n-1} y^{(\mu)}(0) \frac{t_{k+1}^\mu}{\mu!} \frac{1}{\Gamma[\lambda]} \left[\sum_{\mu=0}^k \beta_{\mu,k+1} g(t_\mu, y_\mu) \right]. \quad \dots(28)$$

This implies that $y_{\mu+1}^p$ as per the Adams-Moulton method is represented as:

$$\int_0^{t_{k+1}} [(t_{k+1} - u)^{\lambda-1} g_{k+1}(u)] du = \sum_{j=0}^{k+1} \beta_{\mu,k+1} g(t_j),$$

where

$$\dots(27)$$

$$\beta_{\mu,k+1} = \frac{S_t^\lambda}{\lambda} [(k+1-\mu)^\lambda - (k-\mu)^\lambda]$$

And the predictor form, $y_{\mu+1}^p$ of this fractional Adams-Moulton method can be seen as:

$$y_{k+1} = \sum_{\mu=0}^{n-1} y^{(\mu)}(0) \frac{t_{k+1}^{\mu}}{\mu!} \frac{1}{\Gamma[\lambda]} \left[\sum_{\mu=0}^k \alpha_{\mu,k+1} g(t_{\mu}, y_{\mu}) + \alpha_{\mu+1,k+1} g(t_{\mu+1}, y_{\mu+1}^p) \right]. \quad \dots(26)$$

$$\int_0^{t_{k+1}} [(t_{k+1} - u)^{\lambda-1} \tilde{g}_{k+1}(u)] du = \sum_{j=0}^{k+1} \alpha_{\mu,k+1} g(t_{\mu}),$$

where

...(25)

$$\alpha_{\mu,k+1} = \frac{S_t^{\lambda}}{\lambda(\lambda+1)} \left\{ \begin{array}{l} k^{\lambda+1} - (k-\lambda)(k+\lambda)^{\lambda}, \mu=0 \\ (k-\mu+2)^{\lambda+1} + (k-\mu)^{\lambda+1} - 2(k-\mu+1)^{\lambda+1}, 1 \leq \mu \leq k \\ 1, \mu = k+1 \end{array} \right\}$$

having $g(u)$ as the piecewise linearly inclined interpolation polynomial with nodes calculated through $\tilde{g}_{k+1}$, $t_j = j S_t$. Thus, according to the standard quadrature rules, the RHS of the eqn. (24) can be represented through:

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$$\int_0^{t_{k+1}} [(t_{k+1} - u)^{\lambda-1}] g(u) du \approx \int_0^{t_{k+1}} [(t_{k+1} - u)^{\lambda-1} \tilde{g}_{k+1}(u)] du, \quad \dots(24)$$

Now, apply trapezoidal rule with the weight function $(t_{k+1} - u)^{\lambda-1}$ on the eqn. (13) which results in the following:

$$y_{k+1} = y_k + \frac{S_t}{2} [g(t_k, y_k) + g(t_{k+1}, y_{k+1}^p)] \quad \dots(23)$$

$$y_{k+1}^p = y_k + S_t f(t_k, y_k) \quad \dots(22)$$

$$\int_{\alpha}^{\beta} [h(u)] dz \approx (\beta - \alpha)[h(\alpha)] . \quad \dots(21)$$

Clearly, this becomes an implicit form as h_{k+1} appears on both sides of the eqn. (20). Thus, assume predictor term y_{k+1}^p for h_{k+1} in the sense:

$$y_{k+1} = y_k + \frac{S_t}{2} [g(t_k, y_k) + g(t_{k+1}, y_{k+1})] \quad \dots(20)$$

and then eqn. (16) becomes: $y(t_k) = y_k$ such as $y(t_{k+1}) = y_{k+1}$.

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$$h(t_{k+1}) = h(t_k) + \frac{S_t}{2} [g(t_k, h(t_k)) + g(t_{k+1}, g(t_{k+1}))]. \quad \dots(19)$$

$$\int_{\alpha}^{\beta} [h(u)] dz \approx \frac{\beta - \alpha}{2} [h(\alpha) + h(\beta)] . \quad \dots(18)$$

$$h(t_{k+1}) = h(t_k) + \int_{t_k}^{t_{k+1}} [g(u, h(u))] du \quad \dots(17)$$

$$h_{\mu} = h(t_{\mu}), \mu = 0, 1, 2, \dots, k \quad \dots(16)$$

$$S_t = \frac{T}{R}, \quad t_\mu = \mu S_t, \quad \mu = 0, 1, 2, \dots, N,$$

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The points in the grid form consisting of every point of eqn. (14) are to be divided by R steps as per the interval $[0, T]$ as:

$$h^{(k)}(0) = h_0. \quad \dots(15)$$

$$D[h(t)] = g(t, h(t)), \quad \dots(14)$$

$$h(t) = \sum_{\mu=0}^{n-1} h^{(\mu)}(0) \frac{t^\mu}{\mu!} \frac{1}{\Gamma[\lambda]} \int_0^t [(t-u)^{\lambda-1} g(u, h(u))] du. \quad \dots(13)$$

$$J^\beta[h(t)] = \frac{1}{\Gamma(\beta)} \int_0^t [(t-u)^{\beta-1} h(u)] du, \quad \beta > 0. \quad \dots(12)$$

$$D_*^\lambda[h(t)] = J^{n-\lambda}[D^n[h(t)]], \quad n > \lambda \quad \dots(11)$$

where $D_*^\lambda[h(t)]$ denotes the operator as defined by the Caputo as:

$$h^{(k)}(0) = h_0^{(k)}, k = 0, 1, 2, 3 \dots, \lceil \lambda \rceil - 1$$

$$D_*^\lambda [h(t)] = g(t, h(t)), \lambda > 0, \quad \dots (10)$$

$$\Gamma(\lambda) = \int_0^\infty \tau^{\lambda-1} e^{-\tau} d\tau$$

having $\Gamma(\cdot)$ as the well-known gamma function computed through:

$${}^{CD}D_{0+}^\lambda h(t) = \frac{1}{\Gamma(n-\lambda)} \int_0^u \frac{h^n(\tau)}{(t-\tau)^{\lambda-n+1}} d\tau, t > 0; n-1 < \lambda < n \quad \dots (9)$$

The integral for the order, $n-1 < \lambda < n$ defined in the sense of Caputo as:

$$I_{0+}^\lambda = \frac{1}{\Gamma(\lambda)} \int_0^u [(t-\tau)^{\lambda-1} g(\tau)] d\tau, t > 0 \quad \dots (8)$$

For $\lambda > 0$; $h \in L^1([0, u], \mathbb{R})$ having $[0, u] \subset \mathbb{R}_+$, the fractional integral of order λ for the function $h(t)$ following the Riemann-Liouville conditions can be calculated as:

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$${}^C D_l^\lambda h(t) = \frac{1}{\Gamma(n-\lambda)} \int_l^u \frac{D^n h(\tau)}{(t-\tau)^{\lambda-n+1}} d\tau, n-1 < \lambda < n \quad \dots(7)$$

Caputo derivative is defined for fractional order, $n-1 < \lambda < n$ as:

$${}^R D_l^\lambda h(t) = \frac{1}{\Gamma(1-\lambda)} \frac{d}{dt} \int_l^u \frac{h(y)}{(t-y)^\lambda} dy \quad \dots(6)$$

Whenever, $0 < \lambda < 1$, the RL operator reduces the equation to:

$${}_l D^{-n} h = \frac{1}{(n-1)!} \int_l^y (y-\tau)^{n-1} h(\tau) d\tau \quad \dots(5)$$

τ – time function

Now, employing fractionalization algo for extending this fractional equation i.e., as $n \rightarrow \lambda$, the

$${}_l D^{-n} h = \int_l^{y_n} \int_l^{y_{n-1}} \dots \int_l^{y_1} h(y_0) dy_0 \dots dy_{n-1}, n \in \mathbb{N} \quad \dots(4)$$

$$\Gamma(\lambda) = \int_0^{\infty} \tau^{\lambda-1} e^{-\tau} d\tau$$

having $\Gamma(.)$ as the well-known gamma function computed through:

$$\begin{aligned} {}^R D_u^\lambda h(t) &= \left(\frac{d}{dt} \right)^n \left({}_l D_u^{-(n-\lambda)} \right) h(t) \\ &= \frac{1}{\Gamma(n-\lambda)} \left(\frac{d}{dt} \right)^n \int_l^u \frac{h(y)}{(t-y)^{\lambda-n+1}} dy, \end{aligned}$$

where, ...(3)

$$\lambda > 0; n = [\lambda] + 1; u > l$$

R – Reimann-Liouville

l – lower bound

u – upper bound

λ –fractional-order

$$\binom{\lambda}{m} = \frac{\lambda(\lambda-1)(\lambda-2)\dots(\lambda-m+1)}{m!} \quad \dots(2)$$

$$D^\lambda h(t) = \lim_{n \rightarrow 0} \frac{1}{n^\lambda} \sum_{m=0}^{\infty} (-1)^m \binom{\lambda}{m} h(t - mn), \quad n \in \mathbb{N} \quad \dots(1)$$

$$\begin{aligned} P(\lambda) = |J - \lambda I| = \\ |-\lambda^3 + \frac{(-\sqrt{R} - \eta V - FV - MV)\lambda^2}{V} \\ + \frac{1}{V} \left(-F\sqrt{R} - M\sqrt{R} + \eta^2 V - \eta FV \right. \\ \left. - \eta MV - FMV - Vx(F+x) - \eta Vz \right) \lambda \\ + \frac{1}{V} \left(-FM\sqrt{R} + \eta^2 MV - \eta FMV - \sqrt{R}x(F+x) \right. \\ \left. - \eta Vx(F+x) - \eta Vxy - \eta MVz \right) | \end{aligned}$$

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$$J = \begin{pmatrix} -eta - \frac{\sqrt{R}}{V} & eta & 0 \\ eta - z & -F & x[t] \\ y & F + x[t] & -M \end{pmatrix}$$

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This mathematically developed system has been constructed on the basis of fractional ordered derivatives and the dynamics of human organs being regulated with the oxygen concentration flow from lungs, heart and bloodstream. All variable-parameters: $R, V, eta, F, M, Q > 0$ are positive constants with the initial conditions: $x(0), y(0), z(0) \geq 0$. The quantities are positive whenever they are added to the system and the quantities leaving the system are taken to be negative. The nonlinear techniques such as the phase-space simulations have been analyzed. With the help of progression of time, the evolution dynamics of the system with time togetherwith varying characteristics. Analyzations from timeseries help the exploration of structure dynamics that may lead to randomness in the system.

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$${}^C D^{f_1} x = -\frac{\sqrt{R}}{V_i} x - \eta(x - y) + RQ$$

$${}^C D^{f_2} y = x(\eta - z) - Fy$$

$${}^C D^{f_3} z = xy - M_r z + Fy$$

$$\forall x, y, z \in \mathbb{R}^+$$

where

${}^C D^{f_n}$ – Caputo fractional derivative having order f_n

${}^C D^{f_1} x$ – rate of change of oxygen-levels in lungs

${}^C D^{f_2} y$ – rate of change of oxygen-levels in heart

${}^C D^{f_3} z$ – rate of change of oxygen-levels in bodycells

$$x'[t] = ax - b(x - y) + RC$$

$$y'[t] = x(b - z) - dy$$

$$z'[t] = xy - Mz + dy$$

$$\forall x, y, z \in \mathbb{R}^+;$$

with the initial conditions :

$$x(0), y(0), z(0) \geq 0$$

where

$x[t]$ – oxygen flow concentration during any physical state

$y[t]$ – oxygen flow concentration in heart/bloodstream

$z[t]$ – oxygen flow concentration in bodycells

$$a = \frac{\sqrt{R}}{V}; b = \eta; C = Q; d = f$$

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Declarations

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